

Dheeksha S

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Summary

Software Engineering student with strong foundation in AI/ML, distributed systems, scalable backend engineering, and full-stack development. Experienced in building intelligent applications using machine learning, cloud technologies, and real-time architectures.

Education

Vellore Institute of Technology, Vellore Integrated M.Tech Software Engineering	2022 - 2027 CGPA: 8.26
Ethiraj Matriculation Higher Secondary School Class XII	2021 - 2022 82.5%
Ethiraj Matriculation Higher Secondary School Class X	2019 - 2020 86%

Skills

Programming: Java, Python, C++, C, SQL, JavaScript, TypeScript

CS Core: Data Structures & Algorithms, Object Oriented Programming, DBMS, Operating Systems, Computer Networks

AI/ML: Machine Learning, TensorFlow, Computer Vision, Generative AI, EDA

Backend & Cloud: Spring Boot, Spring Security, REST APIs, WebSockets, JWT, Redis, Prisma ORM, AWS S3, Docker

Web: React.js, Next.js, Node.js, Express.js, HTML, CSS, Tailwind CSS, ShadCN UI

Tools: GitHub, VS Code, Docker, Google Colab, Vercel

Projects

AI Multimodal Diabetes Prediction & Retinal Analysis System Dec 2025
Python, TensorFlow, Streamlit, Random Forest, Transfer Learning

- Built multimodal healthcare AI system combining clinical data and retinal image analysis for diabetes risk prediction.
- Developed Random Forest prediction engine using glucose, BMI, insulin, BP, and age-based health parameters.
- Applied transfer learning techniques for diabetic retinopathy detection and intelligent retinal monitoring.
- Created real-time Streamlit dashboard integrating confidence scoring, AI insights, and unified medical decision support.

Real-Time Inventory Reservation & Concurrent Warehouse Management System May 2026
Next.js, TypeScript, Prisma ORM, PostgreSQL, Tailwind CSS, Vercel

- Built concurrency-safe inventory reservation platform with real-time stock tracking and multi-warehouse synchronization.
- Implemented transactional reservation workflows with automatic expiry, confirmation handling, and race-condition prevention.
- Designed scalable idempotent APIs using Prisma transactions and PostgreSQL for reliable concurrent reservation processing.
- Developed interactive analytics dashboard with live inventory visibility and deployment-ready architecture on Vercel.

CloudSyncX: Distributed File Synchronization & Version Control Platform Oct 2025
Java, Spring Boot, PostgreSQL, Redis, WebSockets, AWS S3, Docker

- Built enterprise-grade distributed file synchronization platform with real-time updates and multi-device file management.
- Implemented intelligent file version control system with rollback support and secure cloud-based historical recovery.
- Designed scalable backend architecture using Spring Boot, REST APIs, WebSockets, and concurrent event-driven processing.
- Developed secure JWT-based authentication system with Redis caching and Dockerized deployment-ready infrastructure.

Research / Innovation

A Neuro-Symbolic Knowledge Engineering Framework for Adaptive Intelligent Systems (Patent Initiated)

- Proposed hybrid neural-symbolic framework for explainable AI and adaptive intelligent automation systems.

Certifications

AWS Certified Cloud Practitioner – AWS [Link](#)

Data Visualization – Infosys Springboard [Link](#)